



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.AT.8

Solve Multiplication and Division Equations

Lesson 7-3 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can solve one-step multiplication and division equations using inverse operations.

Language Objective: I can explain my steps using the words inverse operation, multiply, divide, and isolate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Inverse operation

Multiply

Divide

Isolate

Coefficient

Solution



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

An operation that undoes another, like division undoing multiplication, is an

2

To undo dividing by a number, I both sides by that number.

3

To undo multiplying by a number, I both sides by that number.

4

To get the variable alone on one side is to it.

5

The number multiplied by a variable is the .

6 The value of the variable that makes an equation true is the

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How did you undo the multiplication in $3x = 21$ (or the division in $x / 4 = 9$) to isolate the variable?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐

Inverse operation — Two math actions that undo each other, like $\times$ and $\div$.

Operación inversa — Dos operaciones que se deshacen entre sí, como $\times$ y $\div$.

☐

Multiply — To add equal groups again and again.

Multiplicar — Sumar grupos iguales una y otra vez.

☐

Divide — To split into equal groups.

Dividir — Repartir en grupos iguales.

☐

Isolate — To get the letter by itself on one side.

Despejar — Dejar la letra sola en un lado.

☐

Coefficient — The number multiplied by a variable.

Coeficiente — El número que multiplica a una variable.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I divided / multiplied both sides by ____, so the answer is ____.

Dividí / multipliqué ambos lados por ____, así que la respuesta es ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: In $3x = 21$ the variable is multiplied, so I undo it with division.

Evidence: Students read $3x$ as 3 groups of x , recognize multiplication, and contrast it with addition, knowing they will divide (not subtract) to solve.

Reasoning: This evidence connects the definition of inverse operation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I divided / multiplied both sides by** ____, **so the answer is** ____.

Dividí / multipliqué ambos lados por ____, así que la respuesta es ____.

- **The inverse of** ____ **is** ____, **so I** ____.

El inverso de ____ es ____, así que ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** ____.

Mi afirmación es ____.

- **I know because** ____.

Lo sé porque ____.

- **So** ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Inverse operation
2. **Notes 2:** Multiply
3. **Notes 3:** Divide
4. **Notes 4:** Isolate
5. **Notes 5:** Coefficient
6. **Notes 6:** Solution
7. **Watch for this mistake:** *A common mistake in Solve Multiplication and Division Equations is applying the SAME operation again instead of the inverse — for example, seeing $6x = 42$ and multiplying both sides by 6 to get $x = 252$, instead of dividing by 6 to get $x = 7$. Before you submit, ask: did I use the OPPOSITE operation shown in the equation, and did I apply it to both sides?*
8. **Try It 1:** A. $x = 18$ — x is divided by 3, so multiply both sides by 3: $x = 6 \times 3 = 18$. Check: $18 / 3 = 6$.
9. **Try It 2:** A. $n = 9$ — The 5 is multiplying n , so divide both sides by 5: $n = 45 / 5 = 9$. Check: $5 \times 9 = 45$.